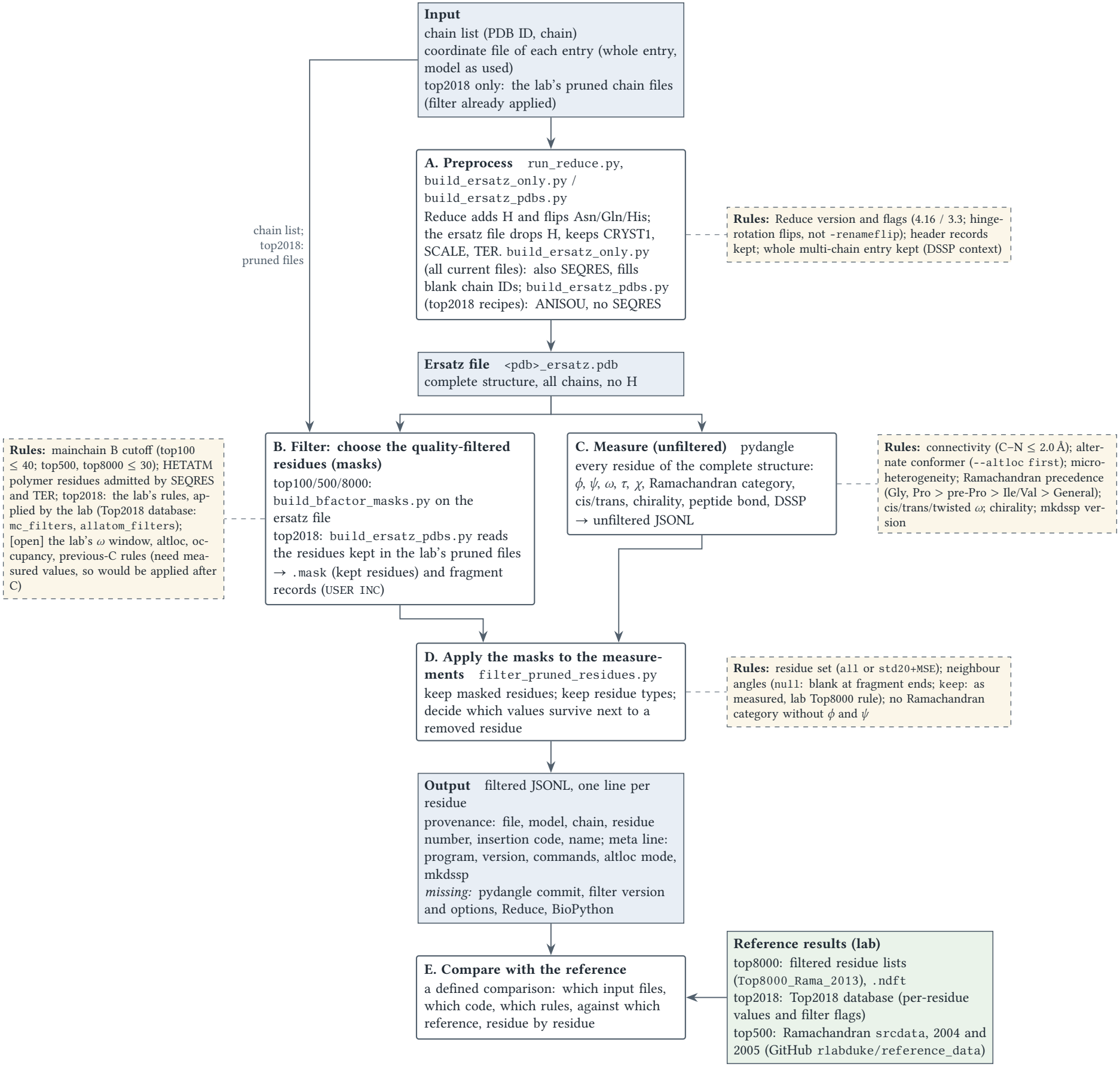
Input
chain list (PDB ID, chain)
coordinate file of each entry (whole entry, model as used)
top2018 only: the lab's pruned chain files (filter already applied)
chain list; top2018: pruned files
A. Preprocess run_reduce.py, build_ersatz_only.py / build_ersatz_pdbs.py
Reduce adds H and flips Asn/Gln/His; the ersatz file drops H, keeps CRYST1, SCALE, TER. build_ersatz_only.py (all current files): also SEQRES, fills blank chain IDs; build_ersatz_pdbs.py (top2018 recipes): ANISOU, no SEQRES
Rules: Reduce version and flags (4.16 / 3.3; hinge-rotation flips, not -renameflip); header records kept; whole multi-chain entry kept (DSSP context)
Ersatz file <pdb>_ersatz.pdb
complete structure, all chains, no H
Rules: mainchain B cutoff (top100 ≤ 40; top500, top8000 ≤ 30); HETATM polymer residues admitted by SEQRES and TER; top2018: the lab's rules, applied by the lab (Top2018 database: mc_filters, allatom_filters); [open] the lab's ω window, altloc, occupancy, previous-C rules (need measured values, so would be applied after C)
B. Filter: choose the quality-filtered residues (masks)
top100/500/8000: build_bfactor_masks.py on the ersatz file
top2018: build_ersatz_pdbs.py reads the residues kept in the lab's pruned files → .mask (kept residues) and fragment records (USER INC)
C. Measure (unfiltered) pydangle
every residue of the complete structure: φ, ψ, ω, τ, χ, Ramachandran category, cis/trans, chirality, peptide bond, DSSP → unfiltered JSONL
Rules: connectivity (C–N ≤ 2.0 Å); alternate conformer (--altloc first); micro-heterogeneity; Ramachandran precedence (Gly, Pro > pre-Pro > Ile/Val > General); cis/trans/twisted ω; chirality; mkdssp version
D. Apply the masks to the measurements filter_pruned_residues.py
keep masked residues; keep residue types; decide which values survive next to a removed residue
Rules: residue set (all or std20+MSE); neighbour angles (null: blank at fragment ends; keep: as measured, lab Top8000 rule); no Ramachandran category without φ and ψ
Output filtered JSONL, one line per residue
provenance: file, model, chain, residue number, insertion code, name; meta line: program, version, commands, altloc mode, mkdssp
missing: pydangle commit, filter version and options, Reduce, BioPython
E. Compare with the reference
a defined comparison: which input files, which code, which rules, against which reference, residue by residue
Reference results (lab)
top8000: filtered residue lists (Top8000_Rama_2013), .ndft
top2018: Top2018 database (per-residue values and filter flags)
top500: Ramachandran srcdata, 2004 and 2005 (GitHub rlabduke/reference_data)